Adnan Shahriar, PhD

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PROFESSIONAL SUMMARY

Ph.D. mechanical engineer developing first-principles simulation tools and workflows for structural, thermal, and fluid systems. Built custom FEM/XFEM solvers, nonlinear material models, and experimentally calibrated structural simulations. Developed C/C++ plant models, Python/CadQuery automation, and CAD-to-mesh algorithms, with additional research in reduced-order methods and simulation-trained damage surrogates. Applied CFD/FEA and prototype testing to mechanical product design.

TECHNICAL SKILLS

Physics-Based Simulation: Custom FEM/XFEM solvers, Nonlinear material models, Thermal-fluid coupling, Structural dynamics

Engineering Software: Python, C/C++, CadQuery, Jenkins CI/CD, Numerical algorithm optimization, Bash, PowerShell, Simulation integration

Geometry and Modeling: Parametric CAD, STEP export, CAD-to-mesh preprocessing, Reduced-order models, Neural-network surrogates

Experimental Validation: Prototype testing, Impact-hammer testing, Structural model calibration

PROFESSIONAL EXPERIENCE

Project Engineer — Rheem Manufacturing (Friedrich Air Conditioning) | San Antonio, TX *Mar 2025–Present*

- **Parametric Fan-Shroud Generation**
  - Developed a Python/CadQuery-based tool that converts fan diameter, coil geometry, component locations, and design clearances into 3D shroud geometry and STEP files, reducing repetitive CAD modeling and accelerating design iteration across product configurations.
- **Engineering Workflow Automation**
  - Built workflows linking parametric design inputs, numerical evaluation, optimization, and CAD output to enable programmatic exploration of engineering alternatives.
- **Motor-Mount Structural Optimization**
  - Redesigned a motor-mount structure using FEA, reducing material usage by more than 50% while maintaining stiffness and vibration requirements. Simplified injection-molded components to reduce secondary manufacturing operations and assembly complexity.
- **Structural FEA and Assembly Optimization of Modular AHU**
  - Performed comprehensive finite element analysis (FEA) of the nine-segment modular AHU assembly to evaluate structural integrity, press-fit connections, and load-bearing capacity. Analyzed deformation under fan and motor loads to maintain critical clearances between the rotating fan lip and stationary bellmouth, preventing mechanical interference and supporting reliable prototype operation.
- **Explicit Dynamic Impact and Fracture Assessment**
  - Performed explicit dynamic FEA impact analysis of plastic components to evaluate impact-induced stresses and potential cracking; calculated the J-integral to assess the driving force for crack propagation.
- **Outdoor Fan-Shroud Optimization**
  - Used CFD to optimize shroud geometry and improve airflow distribution across the condenser coil. Fabricated and tested a prototype, demonstrating more than 80% improvement in airflow efficiency (CFM/W) relative to the previous design.

Senior Simulation Integration Engineer — L&T Technology Services | Plano, TX *Jan 2024–Mar 2025*

- **Real-Time C/C++ HVAC Plant Model for SiL and HiL**
  - Developed a real-time C/C++ executable HVAC plant model for Software-in-the-Loop (SiL) and Hardware-in-the-Loop (HiL) applications, combining finite element solution of transient heat transfer and velocity-pressure-coupled Navier-Stokes equations with lumped-capacitance heater/cooler models, convective thermal-fluid coupling, and a proportional-derivative (PD) HVAC controller model.

- **Modular GUI for Aerodynamic Data Analysis**
  - Developed a modular GUI for aerodynamic data analysis with well-defined input/output interfaces, improving maintainability and reducing computational and memory overhead.
- **Jenkins CI/CD and Automated Debugging**
  - Developed Jenkins CI/CD pipelines with Bash and PowerShell scripting and automated screen-capture capability, streamlining debugging and test workflows for construction and forestry vehicle software.
- **Integrated SiL Simulation Infrastructure and Vehicle Network Virtualization**
  - Developed and integrated a Software-in-the-Loop (SiL) infrastructure connecting the simulator, display, virtual CAN/LIN communication, and ECU-testing components to support construction and forestry vehicle software testing.

Graduate Researcher — Resilient Extraterrestrial Habitats Institute | San Antonio, TX *Dec 2019–Jan 2024*

- **Custom 3D Multiphysics Finite Element Solvers**
  - Developed custom three-dimensional finite element solvers from first principles for structural mechanics, transient heat transfer, and fluid dynamics, including coupled structural-thermal habitat models and interior thermal-fluid simulations.
- **XFEM, Nonlinear Materials, and Damage Modeling**
  - Developed XFEM methods to represent stress fields around three-dimensional voids and implemented nonlinear material behavior incorporating plasticity, strain-rate effects, thermal expansion, and von Mises and Drucker-Prager yield criteria.
- **Efficient Scientific Software and Numerical Algorithms**
  - Developed computational tools emphasizing reduced arithmetic and memory complexity to support novel numerical formulations and simulation workflows; contributed to a publication on algorithm development.
- **XFEM-Informed Neural Network Damage Surrogate**
  - Developed a neural-network surrogate trained on high-fidelity XFEM simulations to approximate localized damage behavior in larger structural models.
- **Experimental Structural Digital Twin and Model Updating**
  - Created a structural digital twin of an experimental habitat by acquiring impact-hammer and accelerometer data, identifying natural frequencies and mode shapes, using Crypton optical geometry measurements, and calibrating a parameterized beam-element model to match identified modal properties and measured acceleration histories.
- **Reduced-Order Modeling and High-Order Elements**
  - Developed modal-coordinate reduced-order models and applied Guyan static condensation; implemented very high-order finite element formulations for acoustic wave propagation associated with micrometeoroid impacts.
- **NASA Modular-Coupled Virtual Testbed (MCVT)**
  - Developed and integrated structural, thermal, and fluid plant models for NASA's Modular-Coupled Virtual Testbed (MCVT), a system-of-systems simulation environment for virtual testing of resilient extraterrestrial habitats.

Other relevant experience

Universal Real-Time Power Conversion LLC | Milwaukee, WI | June 2022–September 2022

- **CAD-to-Mesh Automation Algorithm**
  - Developed a CAD-to-mesh automation algorithm for engineering simulation preprocessing, applying computational geometry methods with an emphasis on reducing arithmetic complexity and automating mesh generation from CAD models.

SELECTED TECHNICAL DEMONSTRATIONS

Finite element formulation software tool: https://www.linkedin.com/feed/update/urn:li:activity:7122914707790843904/

Large-deformation beam simulator: https://www.linkedin.com/feed/update/urn:li:activity:7429372567318331392/

Lunar habitat thermal topology optimization: https://youtu.be/EGVZgg1pb6Q

EDUCATION

Ph.D., Mechanical Engineering — University of Texas at San Antonio, 2023 | GPA: 3.94

B.S., Aeronautical Engineering (Aerospace Major) — Military Institute of Science and Technology, 2015